1. Advanced Log Analysis Report

1. Log Correlation and Pattern Identification

This section documents the correlation between authentication failures and anomalous network traffic, which was utilized to identify a multi-stage attack pattern. By linking Event ID 4625 (Failed Logon) with subsequent high-volume outbound traffic, the SOC identified a successful credential compromise followed by data exfiltration.

1 fx Timestamp

	A	B	C	D	E	F	G	H
1	Timestamp	Event ID	Source IP	Destination IP	Notes			
2	2026-03-27 12:0	4625	192.168.1.100	8.8.8.8	Initial reconnaissance; suspicious DNS query.			
3	2026-03-27 12:0	4625	192.168.1.100	185.15.59.224	Anomaly: 1.5MB outbound transfer post-login failure.			
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Fig.1: Alert triage dashboard categorizing incident priority, CVSS scoring, and MITRE ATT&CK framework mapping.

2. Anomaly Detection Rule Configuration

To proactively detect similar threats, a custom rule was implemented within the Elastic Security SIEM. This rule is designed to flag any outbound data transfer exceeding the 1MB threshold within a one-minute window, which is indicative of unauthorized file transfers or beaconing.

- Rule Name: High-Volume Data Transfer Anomaly
- Rule Logic: network.bytes_out > 1000000
- Schedule: Runs every 1 minute with a 1-minute look-back.
- Action: Generate Critical alert and tag for Tier 2 investigation.

Fig. 1: Multi-Stage SOC Workflow Simulation: Containment, Forensic Hashing, and Attack Generation.

3. Log Enrichment and Geolocation Analysis Utilizing a GeoIP enrichment pipeline, the destination IP 185.15.59.224 was analyzed for geographical context.
Enrichment Summary: Geolocation data identifies the destination IP as originating from a data center in the Netherlands. The IP is not associated with known corporate service providers or authorized business partners. The combination of failed authentication attempts and egress to an offshore IP suggests a highly probable Command & Control (C2) connection or a manual data exfiltration event.

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